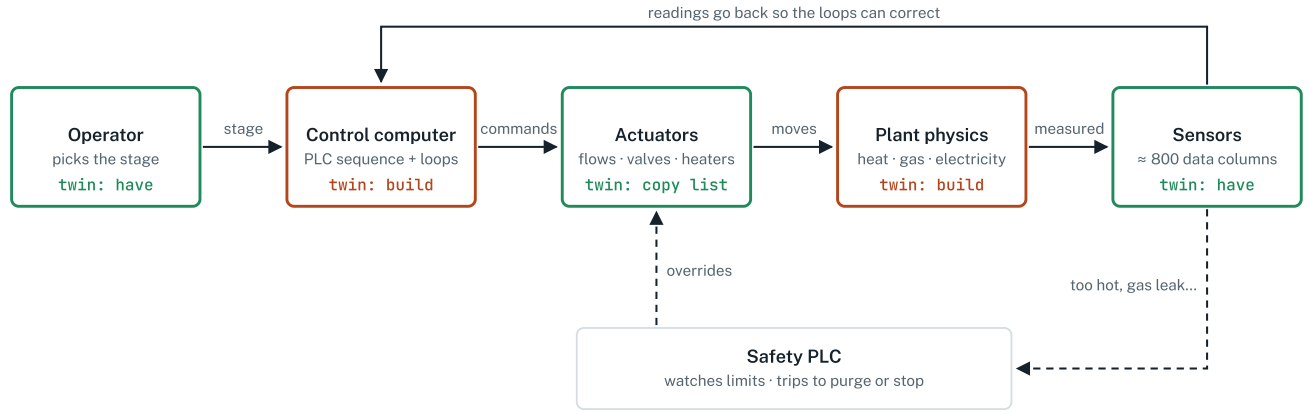
SOFC PLANT · DIGITAL TWIN · 2026-11-03

A twin is the plant's knobs plus the physics in between

In a real SOFC (solid oxide fuel cell) plant, people and the control computer turn only about **20 knobs**: gas flow targets, valves, heater temperature targets and the electric load. The roughly **800 logged readings** are all results of those knobs. A working twin copies the knobs, then uses a simple model of heat, gas and electricity to predict the readings.



▶ Play the loop

Every real plant runs this loop. Green = you already have it for the twin. Orange = still to build. The safety PLC (programmable logic controller) sits outside the loop and can override everything.

What you already have, and what is missing

≈ 20 knobs

Command signals in your data: flow targets, valve commands, heater targets, load settings. Earlier checks showed they **change 1–18 min before** every stage border.

8 stages

Already decided and coded. In the twin they become the **playbook**: for each stage, which knobs are set to what.

Where they fit →

2 min → 4 h

Measured delays from a knob to the readings. The twin must reproduce these. This is the main new work.

See one knob ripple through →

The full knob list →

TWIN PART	STATUS	WHERE IT COMES FROM
Plant layout and sensor positions	have	Your layout and the sensor map
Stages and the sensors in each stage	have	The 8 decided plant stages
Which commands start each stage	have	Command-signal recheck (11 of 11 hot-standby exits had a command first)
How fast each reading follows a command	partly	Response-time page (median delays)
A model that turns commands into readings	build	New: simple heat, gas and voltage blocks fitted to your 39 days
A replay test that proves the model	build	New: feed in recorded commands, compare with recorded readings

> **If you remember one thing:** you do not model 800 sensors one by one. You model about 20 knobs and the few "heat tanks" between them, and the sensors fall out of that.

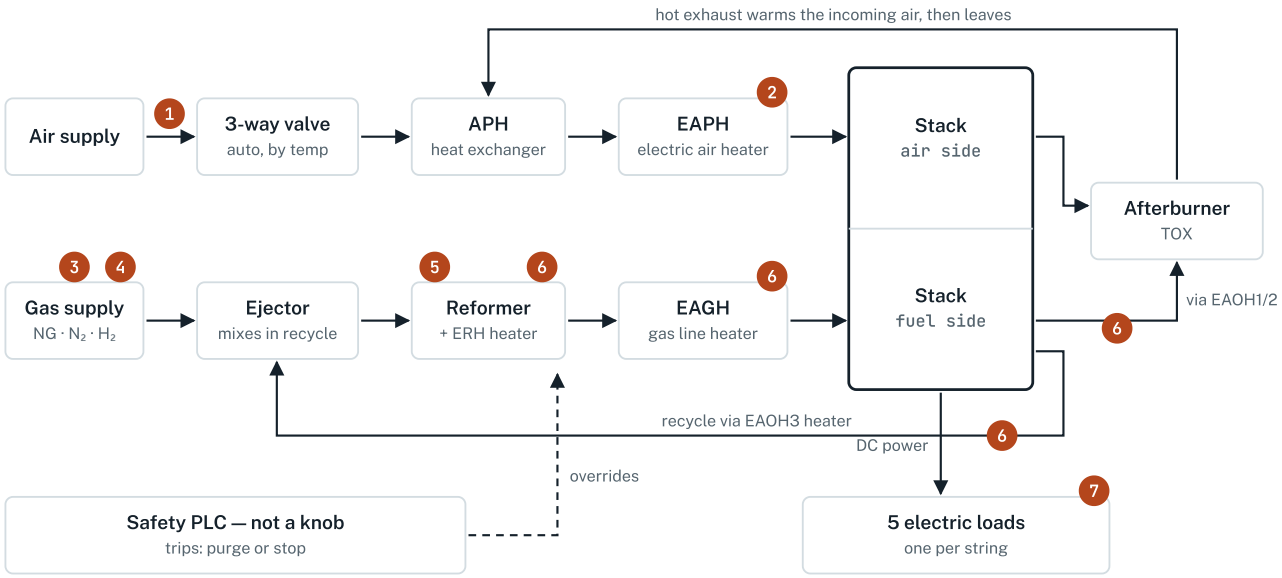
Next: What the plant controls →

Page 1 of 4

PAGE 2 · THE INPUTS OF THE TWIN

What a real SOFC plant actually controls

Seven groups of knobs. Everything else in your data is a reading that follows from them.



Simplified redraw, not to scale. Numbered circles = the knob groups in the table below. The 3-way valve is not a knob: it moves by itself to hold a temperature target.

The knob list

	KNOB	TAG IN YOUR DATA	WHAT IT DOES
1	Cathode air flow target	`FIRC 1,1,1 = 1,1,2`	Main cooling and oxygen supply. Level tells the stage: 6000 normal, 8000–10000 forced cooling. Both controllers get the same target, so it is one knob.
2	EAPH air heater target	`ct_TIRC_206_SP`	The main heat-up knob. Temperature of the air going into the stack.
3	Gas flow targets	`FIRC 1,3 · 1,5 · 1,4`	N_2 and H_2 (forming gas, protects the fuel side when

			hot) and natural gas (the real fuel is FIRC 1,4).
4	Supply valves open/closed	`MV 1,1,2 · 1,2 · 1,3,2 · 1,4,3 · 1,5,3`	Lets each gas in at all. Switches from forming gas to natural gas at fuel switch.
5	POx air target	`FIRC 1,2`	Air mixed with the fuel in the reformer. Set as a ratio to the natural gas.
6	Small heater targets	`TIRC 5,17 · 3,1,1 · 5,3,1 · 5,3,2 · 6,1,1`	ERH, EAGH, EAOH1–3. Keep gas lines hot. They switch as a group.
7	Electric load per string	`EIRC 4.s` on/off, current, ramp	How much current is drawn. Stops itself if a cell falls below 0.75 V.
	Helpers	compressor, exhaust fans	On/off only. Needed, but they do not shape the readings.

Who turns the knobs

LAYER	DECIDES	HOW FAST	IN THE TWIN
Operator	Which stage, and the big targets (for example EAPH 500 → 580 °C)	Minutes to hours	You, using sliders or a scenario script
PLC sequence	Which valves open and which flows run in each stage	Seconds	The stage playbook (from the 8 stages)
Control loops	Heater power and valve opening needed to hold each target	Under a second	A simple "follow the target" rule, no tuning needed at first
Safety PLC	Purge or stop when a limit is crossed (for example stack above 720 °C)	50 ms	A list of trip limits checked every step

What a fuel cell plant tries to hold steady, in plain words: the **stack temperature** (by air flow and air heater), **how much of the fuel the stack uses** (by fuel flow against current), **enough air in the reformer** to avoid soot, **protective gas** on the fuel side whenever the stack is hot and no fuel flows (your rule: above 210 °C), and **slow heating and cooling** so the ceramic does not crack (your heat-ups run at about 1 °C per minute).

> **If you remember one thing:** seven knob groups drive the whole plant. The twin's inputs are exactly this table.

PAGE 3 · WHY THE TWIN NEEDS PHYSICS

One knob moves many sensors, each at its own speed

Turning one knob does not change one reading. It starts a chain through the heater, the air, the stack and the afterburner. A twin has to reproduce both the chain and its delays.

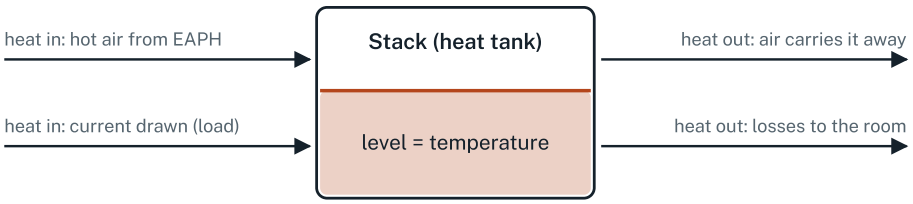


▶ Play the chain

Real example from your data, 2026-03-29, target raised at 13:57. Steps are in order, not to time scale.

Measured delays the twin must match

KNOB CHANGE	FIRST READING THAT MOVES	DELAY (MEDIAN)
Air valve opens	Air flow	1 min
EAPH target up	Air leaving EAPH	2 min
EAPH target up	Stack starts warming · halfway · 90 % there	10 min · 57 min · ≈ 3.6 h
Heaters cut	Stack starts cooling	11 min (6–16)
Natural gas enabled	Stack current	12 min (8–20)
Cold start, heaters on	Stack moves · reaches 300 °C · 500 °C	24 min · ≈ 6 h · ≈ 13.5 h
Load drawn	Stack heats itself	1–2 °C per min



`bigger tank = slower to fill = longer delay`

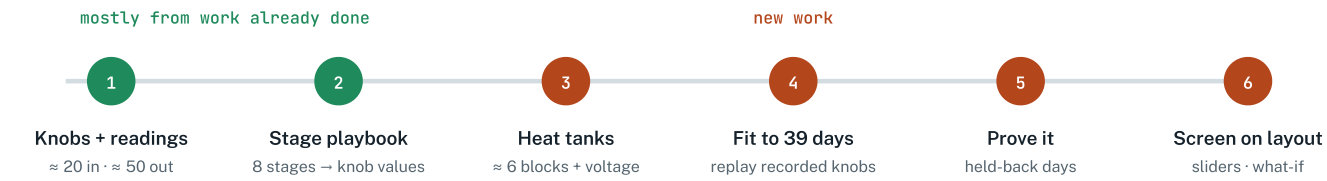
How the twin copies a delay: each hot part is a "heat tank". Its size is fitted so the twin reproduces the delays measured in your data.

If you remember one thing: the delays are the twin. If it gets the 2-minute, 10-minute and 4-hour delays right, the readings will look right.

PAGE 4 · THE PLAN

How to build the twin, step by step

Start with a **grey-box** model: simple physics (heat tanks, gas mixing, stack voltage) whose few numbers are fitted to your 39 days. It is the right size for the data you have.



▶ Play the plan

Six steps. Green steps reuse what you already built. Each step gives something you can check before the next one.

Three ways to build it

	PHYSICS ONLY	DATA ONLY (MACHINE LEARNING)	GREY BOX (RECOMMENDED)
Idea	Full equations of every part	Learn readings from knobs	Few simple equations, numbers fitted to data
Needs	Design data you do not have (stack size, heat loss, materials)	Lots of data in every stage	Your 39 days + the measured delays
Weak spot	Months of work before anything runs	Only 4.5 h of part load; cannot guess 60 kW	Less exact in rare stages; can add a learned correction later

What each step means

1. **Knobs and readings.** Fix the input list (page 2) and about 50 readings to predict (your 50-sensor set).
2. **Stage playbook.** For each of the 8 stages, write which valves are open and which targets are set, plus the switch rules (for example stack above 210 °C → N_2 + H_2 on; trip limits → purge).

3. **Heat tanks.** About six blocks: EAPH, stack, reformer, afterburner, APH, gas-line heaters. Stack voltage = open-circuit voltage minus a resistance times the current, the resistance falling as the stack gets hotter.
4. **Fit.** Feed in the knobs recorded on your 39 days, one step per minute, and adjust the tank sizes until predicted temperatures match the recorded ones.
5. **Prove.** Keep some days out of the fit. Run them and check the error, and check that your stage checker finds the same stage borders.
6. **Screen.** Put the twin behind your layout: sliders for the knobs, parts coloured by temperature, buttons for cold start, load step and trip.

Suggested tools: Python (numpy, scipy) for the model, run one step per minute like your data. A web screen (for example Streamlit or Plotly Dash) for sliders and the layout. OpenModelica only if you later need detailed physics.

How far to take it

LEVEL	WHAT IT DOES	WHEN
Digital model	Runs offline, answers "what if I turn this knob?"	First goal (steps 1–6)
Digital shadow	Follows the live plant data and flags when reality drifts from the model	Next, once the plant streams data
Digital twin	Suggests or sends knob settings back to the plant	Much later, needs site approval

Decisions for you

1. **First scope:** heat only (heat-ups, hot standby, cooling: 905 of your 914 hours) or include power making (only 4.5 h of data)? Suggested: heat only first.
2. **Screen:** which tool is your layout drawn in? That decides how the twin connects to it.
3. **Accuracy target** for the replay test, for example stack within ±20 °C.
4. **Time step:** 1 minute (matches your cleaned data) or 10 seconds (raw data).

> **If you remember one thing:** start with the heat. Steps 1 and 2 are mostly done; step 3 is the first real piece of new work.